MIL-T-3044E
3 September 1976
SUPERSEDING
MIL-T-3044D(ME)
24 July 1969

MILITARY SPECIFICATION

TRUCKS, DUMP, OFF-ROAD, DIESEL-ENGINE-DRIVEN,

15 TO 50 TON PAYLOAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers diesel-engine-driven, four-wheel, rear-wheel drive (4 x 2) and four-wheel drive (4 x 4), off-road trucks having minimum payload capacities of 15 through 50 tons. The dump bodies are the rear-dump type with SAE struck capacities of 9.5 through 30 cubic yards.

1.2 Classification. The dump trucks shall be of the following sizes, as specified (see 6.2):

Size 15 - 15-ton (minimum) payload capacity.
Size 22 - 22-ton (minimum) payload capacity.
Size 35 - 35-ton (minimum) payload capacity.
Size 50 - 50-ton (minimum) payload capacity.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitaiton for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 3805

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

O-A-548	- Antifreeze/Coolant, Engine: Ethylene Glycol, Inhibited, Concentrated.
GG-G-91	- Gages, Tire Pressure, Self-Contained and Inflator-Gage, Pneumatic Tire.
VV-F-800	- Fuel Oil, Diesel.
ZZ-H-428	- Hose, Rubber: Preformed and Straight; Hose, Preformed: Flexible, Wire-Reinforced, (for the Coolant Systems of Automotive and Other Liquid-Cooled Engines).
ZZ-H-461	- Hose, Rubber, Gas (Acetylene-Hydrogen, Air and Oxygen).
ZZ-H-500	- Hose, Rubber, and Hose Assemblies, Rubber; Pneumatic (Yarn or Fabric Reinforced).
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of,
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-G-3787	- Glass Laminated, Flat; (Except Aircraft).
MIL-C-4109	- Coupling Assembly, Low Pressure Air Hose, Quick-Disconnect.
MIL-L-6085	- Lubricating Oil: Aircraft Instrument, Low Volatility.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-L-10324	- Lubricating Oil, Gear, Sub-Zero.

Source: https://assist.dla.mil -- Downloaded: 20260-16T16:04Z
Check the source to verify that this is the current version before use.

MIL-G-10924	- Grease, Automotive and Artillery.
MIL-A-11755	- Antifreeze, Arctic-Type.
MIL-C-21567	- Compound, Silicone, Soft Film.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers, and Trailer Dollies; Preservation and Packaging of.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1224	- Visual Inspection Guide for Pneumatic Tires (Nonaircraft).
MIL-STD-1410	- Methods for Selection of Industrial Engines For End Item Application.
MS15001	- Fittings, Lubrication (Hydraulic) Surface Check, 1/4-28 Taper Threads, Steel, Type I.
MS15002	- Fittings, Lubrication (Hydraulic) Surface Check, Straight Threads, Steel, Type II.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS51318	- Headlight: Blackout, 24 Volt, Waterproof.
MS51330	- Stop Light - Taillight, Vehicular - 24-Volt, Blackout Tail, Blackout Stop.
MS52128	- Switch, Vehicular Lights: 24-Volt DC; (Waterproof).

DRAWINGS

ORD

11682345	- Vehicle Receptacle Assembly NATO Intervehicle Power.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

TECHNICAL INFORMATION PACKAGE

(Application for copies should be addressed to US Army Mobility Equipment Research and Development Command, ATTN: DRXFB-H, Fort Belvoir, VA 22060.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

3. REQUIREMENTS

3.1 Description. Dump trucks shall be the heavy-duty, off-road type. The truck shall consist of the following major assemblies:

(a) Truck chassis with cab.
(b) Hydraulic hoist system.
(c) Dump body with integral canopy.

The truck shall be complete with all required components and accessories supplied by the truck manufacturer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

3.1.1 Physical characteristics. The truck shall conform to the physical characteristics shown in Table I. The weights shown in Table I are for the fully serviced and fueled truck with fully enclosed cab, Rollover Protective Structure (ROPS), Falling Object Protective Structure (FOPS), dump body with integral canopy, hydraulic hoist system, and tires, as specified.

TABLE I. Physical Characteristics

Characteristics	Size 15	Size 22	Size 35	Size 50
Minimum empty weight (lbs)	25,000	35,000	52,000	72,000
Maximum empty weight (lbs)	34,000	46,000	66,000	90,000
Maximum height	12' 0"	13' 0"	14' 0"	15' 0"
Maximum overall width	11' 6"	12' 0"	14' 0"	14' 6"
Minimum SAE struck body capacity (CY) (SAE J741)	9.5	14	21.5	30
Maximum loading height	8'10"	9' 6"	11' 4"	12' 6"
Maximum SAE vehicle clearance Circle (SAE J695)	60'	66'	70'	74'

3.2 First article (preproduction model). The contractor shall furnish one or more trucks as specifeid (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce trucks that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government. When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more trucks for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials, components, and general design not specified shall be those used for the same purpose by the contractor on the standard truck of the size specified

Source: https://assist.dla.mil -- Downloaded: 20260-16T16:04Z
Check the source to verify that this is the current version before use.

that he currently sells on the commercial market and shall be subject to all provisions of this specification. Rubber products furnished, including tires, shall not exceed 8 calendar quarters in age from date of cure of the rubber to date of acceptance of the truck by the Government, and so certified by the contractor.

3.4.1 Threaded parts. Threaded parts shall be in accordance with NBS Handbook H28. All hex-head capscrews shall be SAE Grade 5 or 8, and shall be so marked.

3.4.2 V-belts. V-belts shall conform to SAE J636.

3.4.3 Tolerance information. Component tolerances shall be those tolerances listed in the contractor's quality control program. If requested, the contractor shall make the tolerance information available to the Government prior to the start of the preproduction inspection.

3.5 Truck performance and operating conditions.

3.5.1 Operating temperature. Without preheating, the engine shall start within 5 minutes and the truck shall operate under full-load conditions within 15 minutes after engine start in any ambient temperature in the range of minus 25° F to plus 125° F.

3.5.2 Storage temperature. The truck shall not be damaged by storage under the following conditions:

Temperature	Time period
+ 155° F	4 hours
- 70° F	6 hours, no solar radiation

3.5.3 Performance. The truck shall perform loading, hauling, and dumping of maximum payloads without mechanical adjustment or parts replacement except as specified herein, and without malfunction or breakage of components (except lamp bulbs or sealed beam lamps). Material hauled will be primarily shot rock, as well as crushed rock, aggregate, and earth.

3.5.4 Convoy ability. The truck, with 1/2 rated load in the dump body, shall operate in convoy without malfunction, safety hazard, damage, or permanent deformation, and at an average speed of not less than 20 mph when tested as specified in 4.6.2.20.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

3.5.5 Fording. The truck shall be capable of fording 30-inch-deep freshwater streams without damage to the truck or its components. The truck shall start and operate during and after fording operations without damage to the truck or its components. After testing as specified in 4.6.2.5, the increase of water in enclosures containing lubricants shall not exceed 1.5 percent.

3.5.6 Rain and wind protection. The truck shall be protected from damage due to all combinations of heavy rain and high wind during both operation and extended outside storage. The truck shall start within 2 minutes after exposure to rain and wind when tested as specified in 4.6.2.4 and shall operate without malfunction of any gage, instrument or electrical component. Water in the air intake filter shall not exceed 8 percent of the fluid capacity of the sediment chamber. There shall be no evidence of water leakage into the exhaust system, crankcase sump or hydraulic tank.

3.5.7 Drawbar pull and structural stress. The drawbar pull of the truck at speeds from 1.0 mph to 25 mph shall be such that, when plotted, no point of the curve formed between the intersecting points of the successive gear ratio curves shall fall below the applicable curve shown in Figure A-1. The stress imposed on any structural member through normal and shock loads shall not cause permanent deformation or excessive wear. The truck shall develop the required pull and speed without exceeding an exhaust smoke level of 4.5 on a Robert Bosch EFAW 68 evaluating instrument.

3.5.8 Towing. The truck, with dead engine, shall be towable at speeds up to 10 mph. When being towed, the truck shall be steerable or shall track the towing vehicle within 1/2 truck width. When towing will damage the truck transmission, means shall be provided for two men to disconnect the power train within 30 minutes using common handtools without cutting or welding. A front- and rear-mounted drawbar-type towing device with captive or locking pin shall be provided on each truck. The drawbars shall be capable of withstanding a horizontal load along the longitudinal axis of the truck equal to two times the applicable maximum draw bar pull requirement shown in Figure A-1 without permanent deformation.

3.5.9 Stability. The truck, loaded with the rated payload, shall negotiate uphill and downhill grades ranging from 0 to 30 percent including stop and-start operations. The truck, loaded with the rated

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

payload, shall negotiate sidehill slopes ranging from 0 to 17.5 percent including stop and start and full-turn operations. When the truck is operating under these conditions, no lubricating or hydraulic system shall be adversely affected.

3.6 Safety and human factors.

3.6.1 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be located, insulated, fully enclosed, or guarded to prevent hazards to operating personnel. All moving parts which are of such nature or so located as to be a hazard to operating personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.6.2 Human factors. The characteristics of the truck shall provide for operation by personnel ranging from the small man, clothed, through the large man, arctic clothed, in accordance with SAE J833 and SAE J925.

3.6.3 Rollover Protective Structure (ROPS) and Falling Object Protective Structure (FOPS). The truck shall be equipped with a Rollover Protective Structure (ROPS) and a Falling Object Protective Structure (FOPS) conforming to SAE J1040 and SAE J231, respectively. A commercial structure(s) certified to the above SAE standards is acceptable without retesting, provided that the certification data is made available to the contracting officer, that no structural modifications are required in attaching the structure(s) to the truck, and that the structure(s) meets the other requirements specified herein. The structure(s) shall not interfere with maintainability, ingress into and egress from the truck, and visibility required in normal operation. The interior of the structure(s) shall not protrude into the "space envelope" defined in SAE J154. Each structure shall have the following information legibly and permanently affixed: (1) Contractor's or vendor's name and address, (2) structure model number, if any, and (3) the make, model or series of equipment the structure is designed to fit.

3.6.4 Sound level.

3.6.4.1 Operator station. The sound level of the truck (excluding horns and warning alarms) at the operator's station shall not exceed 85 dB(A) with a closed cab and 90 dB(A) with the operator's window and vent fully open, when tested in accordance with SAE J919.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

3.6.4.2 Exterior. The exterior sound level of the truck shall not exceed 88 dB(A) when tested in accordance with SAE J88.

3.6.5 Backup alarm. A backup alarm conforming to SAE J994 shall be provided.

3.6.6 Horn. The truck shall be equipped with an electric or air-operated horn. The horn shall have either a hand- or foot-operated control.

3.6.7 Fenders. Front and rear fenders shall be provided and shall comply with SAE J321. When fenders are located so that they may be used as access platforms for servicing or repair of the truck or for entering the cab, step pads with nonskid abrasive or expanded metal surfaces shall be provided and the fenders shall be braced to support a weight of 300 pounds without permanent deformation or damage.

3.7 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair and replacement. Items such as connectors, plugs, and valves shall be standardized as to type, and number of sizes shall be held to a minimum. Disconnect plugs and receptacles, multiple-line connectors in the electrical system, and detachable- and attachable-type fittings in the pneumatic system shall be utilized, where applicable, to permit component removal and replacement. Drainage of coolant and lubricants shall be to the ground without draining on any part of the truck. When drains are located over other parts of the trucks, integral tubes, pipes, or other means shall be used to convey the coolant or lubricant from the drain to the ground. Drains shall be accessible without the removal or adjustment of accessories or parts other than opening covers. Each operation in the following maintenance operations list shall be accomplished by one man in not more than the time specified, using special tools furnished with the truck and common tools:

(a) Remove, replace and adjust all engine-driven-belts - 1 hour.
(b) Remove and replace alternator - 1/2 hour.
(c) Remove and replace regulator - 1/2 hour.
(d) Remove and replace steering hydraulic hoses when applicable - 1 hour.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the currentversion before use.

(e) Remove and replace all filters, screens and strainers in all hydraulic systems - 2 hours.
(f) Remove and replace engine coolant system hoses - 1 hour.
(g) Drain engine lubricating oil, remove and replace oil filter elements, and refill crankcase - 1 hour.
(h) Remove and replace fuel filter elements - 1/2 hour.
(i) Disconnect battery cables, remove and replace batteries, and reconnect battery cables - 1/2 hour.
(j) Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers, and refill converter and transmission when applicable - 1-1/2 hours.
(k) Remove and replace starter - 1 hour.

3.8 Drive train. The truck shall be equipped with a drive train consisting of an engine, torque converter, full-torque power-shift transmission, drive shafts with universal joints, planetary drive axle assemblies, and wheel and tire assemblies. Based on a tire-to-ground coefficient of friction of 0.9 and with the torque converter matched to the engine power, each drive train component (including retarder, if in the train) shall be in accordance with the component manufacurer's recommendations for the intended application. This shall include static and dynamic loads and the installation on the truck. Torque ratings shall be the ratings previously published by the manufacturer for commercial use of the component, or if the component is other than a standard commercial item, the rating shall be established by calculations in a manner identical to the manufacturer's previous rating of comparable commercial items. Drive train fluid and lubricant temperatures shall not exceed 250° F when the truck is operated under the conditions and ambient temperatures specified herein.

3.8.1 Diesel engine. For the purpose of selecting the engine, the truck shall be classified as a Class II end item in accordance with MIL-STD-1410. The engine shall be diesel. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, Class II, except as otherwise specified herein. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. The engine shall comply with the Environmental Protection Agency and the California Motor Vehicle Code Regulations governing control of air pollution for new motor vehicles and new motor vehicle engines, in effect on the date of manufacture. The engine shall meet all operational requirements and conditions specified herein without exceeding any of the operating limits specified in Table I of MIL-STD-1410.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

3.8.2 Cranking system. The cranking system shall consist of remote driver controls, starting motor and flywheel-drive mechanism. All components of the system shall be heavy duty with a capacity to crank the engine at a sufficient speed and time interval to insure starting at the lowest ambient temperature specified in 3.5.1. The drive mechanism shall operate and release freely without the application of heat. An electric cranking system shall be furnished on trucks Sizes 15, 22, and 35; an air cranking system shall be furnished on Size 50. For the electric system, the motor shall be a 24-volt type with a solenoid actuated by a starter switch on the instrument panel. The air cranking system shall be the contractor's standard system supplied on equivalent size commercial trucks. In addition, an accessible, permanently mounted, air-brake coupling with shutoff cock shall be provided to permit slave charging of compressed air from another vehicle. The cranking system shall be protected by an automatic starter disconnect system. This system shall automatically de-energize the starter or starter solenoid when the engine starts, and shall prevent starter engagement when the starter switch is activated while the engine is running.

3.8.2.1 Neutral start. An attachment or control mechanism shall be provided within, or connected to, the shift linkage to allow starter operation only when the shift is in the neutral position.

3.8.2.2 Low temperature starting system. A low temperature starting and priming system in accordance with MIL-STD-1410 shall be furnished and shall be electrically (solenoid) operated. The priming system switch shall be located on the cab instrument panel and shall be clearly and permanently marked as to its function.

3.8.3 Air induction system. The engine shall be equipped with a heavy-duty air cleaner and a restriction indicator in accordance with MIL-STD-1410. The restriction indicator shall be mounted where it is visible to the driver in his normally seated position and shall be clearly and permanently marked as to its function.

3.8.4 Exhaust system. The engine exhaust gases shall be routed within the truck body to preclude the payload material from freezing to the body. A valve shall be furnished to direct the exhaust gases to the ground or through an exhaust stack above the cab when the body is in the dump position. An exhaust muffler may be used to meet the requirements of 3.6.4. The exhaust back pressure of the installed system shall

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

not be in excess of the maximum pressure recommended by the engine manufacturer.

3.8.5 Fuel tank. Except as specified herein, a fuel tank conforming to the contractor's commercial standard shall be provided. All tanks shall be provided with a hand-operated fuel shutoff valve attached directly to the tank, and a drain valve which is accessible. The fuel pickup point shall be not less than 1/2 inch above the bottom of the fuel tank. The tank and all lines shall be located or positioned to eliminate fire hazard caused by spilling, overflow, or drainage of fuel. The capacity shall be sufficient for not less than 10 hours of normal operation. The filler neck, strainer, filler-neck cap, and filler-neck-cap gasket may be commercial. The filler-neck opening shall be not less than 3 inches in diameter and shall be located so that: (1) The tank can be filled from a 5-gallon can conforming to MIL-C-1283; and (2) spills and overflow will not come in contact with the engine, engine exhaust system, or starting system. The filler-neck cap shall be captive chained to the tank and shall incorporate a filter to prevent dust from entering the tank and shall incorporate a filter to prevent dust from entering the tank when the tank is vented through the cap. A self-sealing metal bayonet type dipstick, calibrated in increments of 1/4-tank capacity, shall be provided to measure fuel level. The dipstick surface shall be etched, knurled, sandblasted, or phosphate coated to facilitate determining the fuel level. A separate dustproof receptacle for the dipstick is acceptable. A fuel gage is acceptable in lieu of the dipstick.

3.8.6 Torque converter and transmission. The torque converter shall withstand up to 30 seconds of wide-open throttle stall operation without malfunction or damage. The full torque power shift transmission shall provide not less than four forward and one reverse gear ratios for Size 15 and not less than five forward and one reverse gear ratios for Sizes 22, 35 and 50. Forward and reverse gears may be manually selected. A manual or automatic lockup feature shall be provided. All transmission gears shall be constant mesh. The transmission shall provide smooth shifting without slippage that could damage the clutches. Oil shall be filtered by a full-flow filter with a replaceable element having a 10 to 25 micron nominal filter rating. Oil temperatures in the torque converter and transmission shall not exceed 250° F when measured at the torque converter discharge under all operating conditions specified herein. The full power-shift transmission shall permit

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

shifting under full engine power through successive forward and reverse gear ratios. The installation of a downshift inhibitor shall permit upshift and downshift when the transmission is using the converter for multiplication of available torque, and permit upshift and downshift when the torque converter multiplication is not required, including converter lockup.

3.8.7 Retarder. A hydraulic retarder or sealed, fluid-cooled foundation brake assemblies shall be furnished to assist the service brakes to slow the vehicle on curves and downgrades. In all drive ranges, the system shall function as a retarder only when the cavity around the rotor is filled or being filled with oil. A control valve that directs oil into the cavity when retarding is desired will be actuated by the driver's left foot. Oil-cooled disc brakes, capable of continuous operation as a retarder to assist the service brakes and operated by a separate control valve located convenient to the operator, may be substituted for the hydraulic retarder.

3.8.8 Cooling system. Coolant hose, with the exception of the radiator hose, shall conform to ZZ-H-428, Type I, Grade B, Class 4. The radiator hose shall be commercial standard, capable of withstanding coolant temperatures of not less than 210° F.

3.8.9 Differential. When a four wheel, rear wheel drive (4 x 2) truck is furnished, a no-spin, limited-slip, torque proportioning, torque bias, or high traction type differential shall be provided on the rear axle. With poor underfoot conditions, the maximum percentage of torque delivered to the wheel with better traction shall be not less than 150 percent of the torque delivered to the wheel with poor traction.

3.9 Brake systems. A service brake system, an emergency stopping system, and a parking system shall be provided. The brake systems may use common components.

3.9.1 Service brake system. Service brakes shall be provided on all four wheels and shall be air or air-over-hydraulic actuated. Service brakes shall be foot controlled. The brake pedal shall be positioned for control by the operator's right foot. The brakes shall safely bring the truck to a complete stop within the distance specified in Table II from the instant of brake pedal application from a speed of 20 mph at

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

rated GVW. All wheels shall remain in contact with the pavement when tested as specified in 4.6.2.6. Heating of the brakes by repeated stops shall not adversely affect the stopping distance. The air system shall be equipped with a pressure gage and a low-pressure audio warning device. The air system shall be equipped with an alcohol evaporator to prevent system freezing.

3.9.2 Emergency stopping system. The truck shall be equipped with an emergency stopping system. The emergency stopping system shall safely bring the truck to a complete stop in the event of any single failure in the service brake system. The stop shall be within the distance specified in Table II from the instant of brake application from a speed of 20 mph at rated GVW. All wheels shall remain in contact with the pavement when tested as specified in 4.6.2.7. The emergency system shall be capable of being applied manually by a person seated in the operator's seat. The system shall be arranged so that it cannot be released from the operator's seat after any application unless immediate reapplication can be made from the operator's seat to stop the truck. In addition to the manual control, the emergency system may also be applied automatically. If an automatic emergency stopping system is used, the automatic application shall occur after a warning device is actuated.

TABLE II. Brake Performance Requirements

Truck Size	Service Brake Maximum Stopping Distance from 20 mph (ft)	Emergency Stopping System Maximum Stopping Distance from 20 mph (ft)
15	45	135
22	50	150
35	60	180
50	65	195

3.9.3 Parking system. A parking system shall be provided and shall be capable of holding the truck loaded with the rated payload on a 30 percent longitudinal dry-swept concrete grade in either direction. The parking system shall be controlled by a hand or a foot control accessible to the operator. The parking system control shall not require more than a 90-pound push or pull to meet the above requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

3.9.4 Air supply coupling. The truck shall be equipped with an air supply coupling to provide a connection for the inflation of tires from the brake reservoir delivery line. The coupling shall be labeled "air supply" in accordance with 3.21.2, and shall conform to MIL-C-4109, Type I, Class A, Style 2, Size 3/8. When specified, a commercial standard coupling may be provided (see 6.2).

3.10 Tires. All tires shall be tubeless and shall have the rock-type (E-3) tread pattern. All tires shall be the same size and type. All tires shall have the same ply rating or load range. Commercial anti-ozonants shall be included in the tires during manufacture. Tires shall have no visible defects when examined in accordance with MIL-STD-1224. With the truck loaded to rated capacity, the tire loads shall not exceed 105 percent of the tire load limit recommendations of the TRA Yearbook for a 30 mph rating speed. Minimum tire sizes and minimum ply ratings and the number of tires required per truck shall be as listed in Table III. Radial ply tires may be furnished.

TABLE III. Tire Size and Ply Ratings

Truck	Minimum Tire Size	Minimum Tire Ply Rating	Number Required
15 (4 x 2)	14.00 x 25	20	2 front, 4 rear
22 (4 x 2)	16.00 x 25	28	2 front, 4 rear
35 (4 x 2)	18.00 x 33	32	2 front, 4 rear
35 (4 x 4)	18.00 x 25	24	4 front, 4 rear
50 (4 x 2)	21.00 x 35	36	2 front, 4 rear
50 (4 x 4)	18.00 x 33	32	4 front, 4 rear

3.10.1 Wheels. Dual spacing and rim width shall conform to the TRA Yearbook.

3.10.2 Valve parts. All valve parts shall conform to the TRA Yearbook.

3.10.3 Tire inflator. When specified (see 6.2), a tire inflator conforming to GG-G-91, Type III, Style 3, with the following exceptions, shall be furnished:

(1) The discharge hose between the inflation valve body and the foot chuck coupler shall be 5 feet in length.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(2) The tire valve adapter assembly shall consist of one adapter to screw onto standard (0.302-32) thread size, one adapter to screw onto off-highway tire valves (0.482-26) thread size, tire valve extension, clip, ring, and chains. The inflator shall have a leader hose of sufficient length to reach all tires. The hose shall conform to ZZ-H-461 or ZZ-H-500 and shall be fitted with air-brake-line couplings for mating to the coupling furnished in accordance with 3.9.4.

3.11 Hoist system. The hoist system shall consist of components rated to perform the hydraulically powered operations of the dump body with the specified payload. Mechanisms and reservoirs shall be positioned or guarded to prevent any possible hazard or damage from hot surfaces, debris or obstructions. The operating controls for the hoist system shall be located inside the cab, accessible to the operator. The dump-hoist control shall be the 4-way type providing hoist, lower, float, and hold. No damage or shock shall result from the control being held in the "full hoist" position up to and at the maximum dump angle, or in the "full lower" position down to and at the fully down position. The hoist system shall include a pressure-relief valve and a full flow replaceable filter. Oil for the system shall conform to MIL-L-2104, Grade 10, or MIL-L-46167. All components and materials shall be compatible with these oils and shall operate as specified herein throughout an oil operating temperature range of minus 25° to 225° F. For low-temperature operation, oil conforming to MIL-L-46167 may be used. The temperature rise of the oil in any portion of the hydraulic system shall not exceed 100° F in ambient temperatures of 50° F and higher during all operations specified herein. The maximum working pressure of the hydraulic system shall not exceed 2500 pounds per square inch (psi).

3.11.1 Reservoir. The reservoir shall be mounted to prevent damage due to vibration. The filler neck shall contain a strainer retained in place. Except when a closed system is provided, an air-breather with a 100-mesh or finer filter shall be furnished. A magnetic drainplug or doughnut-type magnet, accessible for inspection and cleaning, shall be provided in the reservoir. The reservoir shall be designed to minimize agitation or aeration of the oil. A sight gage or self-sealing dipstick with "ADD" and "FULL" marks on one side shall be furnished for checking the oil level, and shall be coated and scored to clearly show the oil level. The dipstick shall be furnished with a threaded cap and gasket seal. All surfaces inside the reservoir which are exposed to the reservoir oil shall be permanently treated to prevent corrosion and the formation of other types of contaminants.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the currentversion before use.

3.12 Steering system. The truck shall be equipped with a fully hydraulic or hydraulic-assist power steering system for the front wheels which shall provide precise steering control of the truck at all speeds and during all operations specified herein. The steering system components shall be matched to the rated load capacity of the front axle as recommended by the manufacturer of the front axle, and shall be intended for heavy-duty, off-road, dump-truck operation. The steering system shall enable the operator to safely guide the truck with a dead engine to a stop off the roadway from a speed of not less than 25 mph. The steering shall be controlled by a steering wheel. The effort required to turn the steering wheel shall not exceed 18 pounds pull at the rim with the engine running at low idle when the dump truck is not in motion. The steering system shall have no components in common with the hoist system.

3.13 Electrical system. Unless otherwise specified (see 6.2), each truck shall be equipped with a 24-volt electrical system.

3.13.1 Wiring. Wiring and harness shall be clamp supported, and no unsupported length shall exceed 3 feet. All wiring shall be located to avoid damage. Wiring shall be identified in accordance with SAE J210. Grommets shall be used to prevent chafing where wires pass through holes in structural members and other metal parts.

3.13.2 Slave receptacle. A slave receptacle conforming to ORD Drawing 11682345 shall be installed near the batteries to permit engine slaving and battery charging.

3.13.3 Lighting. Unless otherwise specified (see 6.2), the truck shall be equipped with not less than four sealed beam headlights conforming to Trade No. 4880 of SAE J598. Two taillights and two stoplights shall be provided in accordance with SAE J585 and SAE J586.

3.13.4 Blackout lights. One blackout headlight conforming to MS51318 and two blackout stoplight-taillights conforming to MS51330 shall be provided. The blackout headlight shall be mounted on the left front of the truck.

3.13.5 Instrument panel lights. All instruments, gages, and switches on the instrument panel shall be illuminated for night operations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

3.13.6 Light switch. A light switch conforming to MS52128 shall be supplied and shall control the headlights, taillights, blackout lights, and instrument panel lights. Energizing any lights except blackout lights shall not be possible when the light switch is in the blackout position.

3.13.7 Light guards. All lights not recess mounted shall be provided with guards. The guards shall protect the lights from damage from falling debris, tree limbs, and underbrush.

3.13.8 Turn signals. Turn signals shall be supplied and shall comply with SAE J588 and SAE J589. A hazard warning system shall be supplied and shall comply with SAE J945 and SAE J910.

3.13.9 Reflectors. Two red reflex reflectors conforming to SAE J594 shall be installed, one on the right rear and one on the left rear of the truck.

3.13.10 Circuit breakers. All circuits shall be protected by circuit breakers.

3.13.11 Batteries. Batteries shall be in accordance with MS35000, 6TN. The quantity of batteries shall be as recommended by the engine manufacturer for the environmental conditions specified herein.

3.13.11.1 Battery enclosure. A weathertight enclosure of the box type or compartment type shall be provided to contain and protect the batteries. The enclosure shall hold the batteries securely and shall provide accessibility for servicing and for removal of the batteries. The enclosure shall provide clear vision of the electrolyte level in all cells of the batteries. The enclosure shall include drains for removal of water or electrolyte spilled within the container and gas venting at or near the top of the container. The battery enclosure shall be steel with a nominal thickness not less than 0.078 inch and shall be treated with an acid-resistant coating on the interior.

3.13.12 Alternator. A waterproof alternator in accordance with SAE J180 shall be provided. The alternator shall have not less than a 50-amp capacity. The alternator charging system shall be provided with a device to protect the system components from inadvertent reverse polarity and high-current transients resulting from operating with batteries disconnected, with batteries connected in reverse polarity, with dead or shorted cells in a battery, breaking the charging circuit during operation, or connecting leads to the wrong terminals.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

3.13.13 Electromagnetic interference. The electromagnetic interference emission characteristics of the truck shall conform to MIL-STD-461, Class IIIc. Interference caused by starting motors and switches associated with starting, short-duration starting aids, and protective warning devices do not require test.

3.14 Cab. Unless otherwise specified, the cab, mounting method, and cab accessories shall be commercial standard. The cab shall be fully enclosed and shall be fabricated of steel. The cab shall be completely weather-stripped and cold and sound insulated. Transfer of excessive heat into the cab through panels or controls from the engine, torque converter or transmission shall be prevented. Fresh air ventilators shall be provided to direct air to both driver and passenger. A dome light shall be provided. Windshield glass shall be laminated safety-plate glass conforming to MIL-G-3787, Class I, Type A. Other windows shall be of safety glass conforming to MIL-G-3787, Class II. The windows on each side of the cab shall have a means for opening. Interior sun visors shall be furnished to reduce glare through the windshield and shall not be affected by vibration. Two "West Coast" type rearview mirrors shall be furnished and shall be installed so that the operator has a rear view of each side of the truck. The mountings shall prevent vibration from blurring the image of either mirror.

3.14.1 Seating. An upholstered, cushioned operator's seat and backrest shall be provided. The seat shall be of the heavy-duty, air or torsion-spring type having at least a 4-way (front and rear, up and down) seat adjustment and an operator's weight adjustment. Operator space envelope shall conform to SAE J154. The seat shall comply with the requirements of SAE J899. Seat and control locations shall conform to SAE J898. An operator's seatbelt shall be provided and shall conform to SAE J386. An upholstered, cushioned passenger's seat and backrest shall be provided. A passenger's seatbelt shall be provided and shall conform to SAE J386.

3.14.2 Heater and defroster. Unless otherwise specified (see 6.2), a heater and defroster shall be furnished. The heater shall be the hot-water type and shall provide a choice of fresh or recirculated air heating. The fresh-air inlets shall be screened. A forced air duct system for windshield defrosting from the heater shall be supplied. A separate hot-water type defroster may be furnished in lieu of the duct system. Not less than two separate defroster fans shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

furnished and shall be adjustable to aim toward the windshield and side windows. The heater shall maintain a temperature of not less than 35° F within the cab in an ambient temperature of minus 25° F. The heater shall be rated, as installed, at not less than 25,000 Btu.

3.14.3 Windshield wipers and washers. Heavy-duty windshield wiper blades, arms and motors shall be furnished for each major windshield area. One electric or air motor shall be required for each blade and arm assembly. A heavy-duty washer system shall be furnished.

3.14.4 Boarding provisions. Nonslip steps and grab irons for personnel access to normal operating and regular servicing positions shall be provided. Step configuration shall minimize accumulation of mud and snow and shall be suitable for arctic-type boots. The clear opening inside the grab irons shall measure not less than 3 inches by 6 inches.

3.14.5 Instrument panel and instruments. The truck shall be equipped with all instruments, indicators and controls required for the safe and reliable operation of the truck. As a minimum, the following instruments and indicators shall be supplied and mounted in the instrument panel:

(a) Tachometer.
(b) Speedometer with odometer.
(c) Engine oil pressure gage.
(d) Low engine oil pressure warning light.
(e) Engine coolant temperature gage.
(f) High engine coolant temperature warning light.
(g) Ammeter.
(h) Torque converter oil temperature gage.
(i) High torque converter oil temperature warning light.
(j) Transmission oil pressure gage.
(k) Air pressure gage.
(l) Low air pressure warning buzzer.
(m) Fuel level gage or dipstick (see 3.8.5).

Instruments, controls, and wiring shall be accessible for ease of maintenance. Before the engine starts, the actuation of a switch shall illuminate all warning-light circuits. All instruments, gages, and switches in the instrument panel shall be illuminated for night operations. Under sunlight conditions, all warning-light indicators shall be visible to the operator. Temperature and pressure gages shall have luminous

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

numerical markings and shall have the normal operating range and the danger ranges for the particular application marked on the gage faces. The instrument panel shall be fitted with a dispatch box and an ashtray. These shall be installed either integral with the panel or below the panel. The panel shall be removable to provide access to the components. Each indicator and switch shall be identified by a plate conforming to 3.21.2 mounted adjacent to the indicator or switch. Each instrument shall be identified either by a plate conforming to 3.21.2 mounted adjacent to the instrument or on the instrument face in accordance with commercial practice. Additional instruments standard with the contractors commercial practice which contribute to safety and dependability of the truck operation shall be furnished. An hourmeter shall be supplied and may be located either in the instrument panel or in the engine compartment in a location to facilitate reading. All instruments shall be weatherproof and dustproof type and of such quality and accuracy and mounted so as to function under all climatic and operating conditions specified herein.

3.15 Dump body. The truck shall be equipped with a rock-hauling rear dump body, load-bearing full cab protector, and hydraulic hoist system. The dump body size shall be within the SAE struck cubic yard capacity range as specified in Table I. Dumping shall be accomplished so that all types of material, including wet clays, will leave the body. Rock-ejector devices shall be attached to the body and placed so as to drop between the rear dual wheels, offering a positive means for removing stones lodged between the tires. The dump body shall be equipped with positive means of support, permanently attached, and capable of being locked in position to prevent accidental lowering of the body while maintenance or inspection work is being done.

3.15.1 Body fabrication. The body shall be fabricated for shot rock hauling use and shall have, as a minimum, the nominal steel thickness listed in Table IV. Alloy steel thickness minimum shall be based on material having not less than 90,000 psi yield strength.

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

TABLE IV. Body Thickness

Size	Minimum Material Thickness (inches)			
	Vertical Front	Sloped Front	Side	Floor
15	5/16	7/16	5/16	1/2
22	3/8	1/2	5/16	1/2
35	3/8	1/2	3/8	3/4
50	1/2	1/2	3/8	3/4

3.15.2 Canopy fabrication. A load-bearing type canopy shall be provided on the front of the dump body. The canopy shall extend the full width of the dump body and shall protect the cab. For half-cabs, the canopy shall protect the cab and the platform beside the cab. The canopy shall be constructed from steel with a nominal thickness of not less than 3/16 inch and a yield strength of not less than 50,000 psi.

3.16 Toolbox. The truck shall be provided with a toolbox made of steel of 0.078 inch nominal thickness (minimum). The toolbox shall have a hinged lid and a hook fastener. The toolbox shall be securely fastened to the truck in a protected position, shall have provisions for drainage of liquids, and shall accommodate the maintenance tools furnished.

3.17 Fuels, lubricants, and coolants. The truck shall be assembled, run-in and tested with the products of the type and grade specified in Table V for the applicable ambient air temperature. The truck shall operate as specified herein when furnished with these products.

Source: https://assist.dla.mil -- Downloaded: 20260-16T16:04Z
Check the source to verify that this is the current version before use.

TABLE V. Service Products

Item	Application	Specification	Type and Grade
1.	Engine:		
	(a) Heavy duty	MIL-L-2104	10, 30
	(b) Arctic	MIL-L-46167	One grade
2.	Fuel:		
	(a) Diesel	VV-F-800	DF-1, DF-2, DF-A
3.	Grease:		
	(a) Automotive	MIL-G-10924	One grade
	(b) High temperature	MIL-G-23827	One grade
4.	Hydraulic-system oil: (excl. brakes)		
	(a) Oil	MIL-L-2104	10
	(b) Oil (low temperature)	MIL-L-46167	One grade
5.	Torque converter-transmission	MIL-L-2104	10
		MIL-L-46167	One grade
6.	Lubricating oils:		
	(a) Universal	MIL-L-2105	75, 80, 90
	(b) Arctic	MIL-L-10324	One grade
	(c) Instrument	MIL-L-6085	One grade
7.	Antifreeze:		
	(a) Ethylene glycol	O-A-548	One grade
	(b) Arctic	MIL-A-11755	One grade
8.	Brake fluid	MIL-L-2104	10

Note: At the option of the contractor, engines in production trucks may be serviced with high output diesel oils prior to delivery. Gear break-in oils or greases may be utilized provided that such lubricants are compatible with those military lubricants which may be added prior to the first normal maintenance change.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

3.17.1 Lubrication fittings. Lubrication fittings shall conform to MS15001, MS15002 and MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operated quick-disconnect fasteners may be removed to provide accessibility. Where the use of pressure lubricating equipment will damage grease seals or other parts, a suitable pressure-release device shall be provided.

3.17.2 Enclosure with integral reservoir of lubricant. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, finger holes, or sight holes to determine the level of the lubricant. Each enclosure shall be equipped with a magnetic drainplug and with a means for filling the enclosure with lubricant. The drainplug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage, except from planetary axle hubs, shall be to the ground when the truck is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground. Accessibility to the drainplug, the filling means, and the lubricant-level checking device shall be obtained without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operated quick-disconnect fasteners may be removed.

3.17.3 Delivery lubrication service. Unless otherwise specified, all components except those for which preservatives are specified in 5.1 shall be lubricated when delivered. The truck shall be serviced with lubricants and grades of military oils designated for use in the temperature range specified. The truck shall be tagged in a conspicuous place to indicate the temperature range and grades of oils and greases used.

3.18 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.18.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.18.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.19 Treatment and painting. The portions of the truck normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A, color as specified (see 6.2), except for the following procedures:

(a) Crankcase sealer - The application of a crankcase sealer as specified in MIL-T-704 on the interior surfaces need not be applied on the cast-iron housings of transmission and torque converters provided the interior surfaces are void of casting defects such as blowholes, excessive porosity, cold shuts, hard spots, shrinks, cracks, or other injurious defects.
(b) Brake protection - Surfaces of steel and iron brake assembly components which are exposed to air shall be protected by treatment conforming to MIL-T-704, Type II, except that the finish coat is not required. Bearing surfaces in brake assemblies such as brake shoe anchor pins, anchor pin bearing holes in brake shoes, cylindrical surface of pistons in brake cylinders, and inside diameter of brake cylinders for the length of the piston contact, shall be coated with a film of silicone compound conforming to MIL-C-21567. A film of silicone compound shall be applied to the lip and yoke hole of each dust boot and to the boot lip groove in the outside diameter at the ends of each cylinder.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the currenversion before use.

(c) The final paint coat on the inside of the dump body may be delayed until the operational test (see 4.6.2.13), is completed.

3.20 Slinging and tiedown provisions.

3.20.1 Slinging provisions. Permanently affixed slinging provisions that enable the truck to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tiedowns when such provisions conform to MIL-STD-209.

3.20.2 Tiedown provisions. Permanently affixed tiedown provisions that permit securing the truck to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209. Tiedown provisions may also be used as slinging provisions when such provisions conform to MIL-STD-209. The truck and the tiedown provisions shall withstand without damage or permanent deformation the dynamic loads applied when tested as specified in 4.6.2.18.

3.21 Identification and instruction.

3.21.1 Identification marking. The truck shall be identified in accordance with MIL-STD-130 on an identification plate conforming to MIL-P-514, Composition C, of Type I, Grade A, Class 1 material. The plate shall be furnished and mounted by the contractor. A shipping data plate shall be furnished and shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material and, in addition, shall indicate the silhouette of the truck in transport position indicating the center of gravity and location and capacity of the lifting and the tiedown attachments. Each plate shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.21.2 Instruction plates. The trucks shall be equipped with instruction plates including warnings, cautions, and diagrams describing any special or important procedures to be followed in assembling, operating or servicing the truck. A plate showing complete lubrication instructions shall be furnished. Plates shall conform to MIL-P-514 Type III, Composition C, of Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts or rivets in a conspicuous protected location. Instruction plates shall be readable by a man standing on the ground.

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

3.21.3 Stenciling. Registration and other markings shall be applied in accordance with MIL-STD-642. The gross weight of each truck shall be stenciled on each side of the truck. The prescribed tire pressure shall be stenciled on the instrument panel or suitable adjacent area and on the sides of the truck near each tire with block-or stencil-type letters not more than 1 inch nor less than 3/4 inch high.

3.22 Workmanship. Truck shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair their operation or serviceability.

3.22.1 Steel fabrication. Steel used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having eroded surface is not acceptable. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true and all sharp edges and burrs shall be removed. Burned surfaces of flame-cut material shall be free of burrs, sharp edges, and slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat treatment is required. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.22.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with commercial practice. All bolts, nuts and screws shall be tight. Devices securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

3.22.3 Welders and welding.

3.22.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective by the particular code:

AWS D1.1, Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

3.22.3.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings.

3.22.3.3 Welding practices.

(a) Preheat of materials being welded and maximum interpass temperature during welding shall be in accordance with the contractor's recommendations.
(b) Tack welds shall be subject to the same quality requirements as final welds.
(c) Work shall be positioned for flat welding whenever practicable.
(d) Procedures and sequences shall be such that distortion and shrinkage will be held to a minimum. When straightening is required, caution shall be exercised to insure that the straightening process does not weaken the part.

3.22.3.4 Weld quality.

(a) The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled, the truck shall conform to the inspection requirements specified in Section 4.
(b) All weldments shall be free of slag flux, weld, spatter, and other impurities detrimental to either the appearance or strength of the weldment.
(c) Undercut in weldments shall be held to a minimum and shall not be more than 1/32 inch deep.

3.23 Technical Information Package. The contractor shall complete and submit a Technical Information Package within the time frame specified (see 6.2). The information shall depict the complete truck and its components exactly as it will be furnished.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined,and tested in accordance with the requirements of referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Preproduction examination.

4.3.1.1 Pretest examination. Prior to testing, the completed Technical Information Package shall be reviewed. Any discrepancies between the completed Technical Information Package, the specification, and the preproduction model shall be cause for rejection. Prior to testing, the truck shall be examined for the defects marked "X" in Column 1 of Table IV. Presence of one or more defects shall be cause for rejection.

4.3.1.2 Posttest disassembly and examination. Upon completion of all tests, the truck shall be disassembled and examined for the defects marked "X" in Column 2 of Table VI and as specified in 4.6.2.24. Presence of one or more defects shall be cause for rejection.

4.3.2 Preproduction tests. Upon successful completion of the examination specified in 4.3.1.1, the truck shall be subjected to the tests marked "X" in Column 1 of Table VII. Failure of any test shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production trucks will be selected at random by the Government from the first 14 trucks being produced by production tooling and will be examined for the defects marked "X" in Column 3 of Table VI and subjected to the tests marked "X" in Column 2 of Table VII to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initially production truck shall not exclude the remaining trucks from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production truck and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all trucks accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary in furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. After successful completion of all tests specified in 4.5.2, each truck shall be examined for the defects marked "X" in Column 4 of Table VI. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each truck shall be tested as specified in Column 3 of Table VII. Failure of any test shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

4.6 Inspection schedule.

4.6.1 Examination.

TABLE VI. Examination schedule

Pre-test 1	Post-test 2	Initial Production 3	Quality Conformance 4	Defect 5		Requirement Paragraph 6
X	-	X	X	101.	Truck does not conform to general description.	3.1
X	-	X	-	102.	Weight not as specified.	3.1.1
X	-	X	-	103.	Dimensions not as specified.	3.1.1
X	-	X	-	104.	Body capacity not as specified.	3.1.1
X	-	X	-	105.	Material not as specified.	3.4
X	-	X	-	106.	Threaded parts not as specified.	3.4.1
X	-	X	X	107.	Drawbars not as specified.	3.5.8
X	-	X	X	108.	Safety devices not as specified.	3.6.1
X	-	X	-	109.	Human factors characteristics not as specified.	3.6.2
X	-	X	X	110.	ROPS not as specified.	3.6.3
X	-	X	X	111.	FOPS not as specified.	3.6.3
X	-	X	X	112.	Backup alarm not as specified.	3.6.5
X	-	X	X	113.	Horn not as specified.	3.6.6
X	-	X	X	114.	Fenders not as specified.	3.6.7
X	-	X	-	115.	Maintenance features not as specified.	3.7
X	-	X	-	116.	Drive train not as specified.	3.8
X	-	X	-	117.	Diesel engine not as specified.	3.8.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TABLE VI. Examination Schedule (Cont'd)

1	2	3	4	5	6
X	-	X	-	118. Cranking system not as specified.	3.8.2
X	-	X	X	119. Neutral start provision not as specified.	3.8.2.1
X	-	X	-	120. Low temperature starting provisions not as specified.	3.8.2.2
X	-	X	X	121. Air cleaner not as specified.	3.8.3
X	-	X	X	122. Exhaust system not as specified.	3.8.4
X	-	X	-	123. Fuel tank not as specified.	3.8.5
X	-	X	-	124. Torque converter not as specified.	3.8.6
X	-	X	-	125. Transmission not as specified.	3.8.6
X	-	X	-	126. Retarder not as specified.	3.8.7
X	-	X	-	127. Cooling system not as specified.	3.8.8
X	-	X	-	128. Service brake system not as specified.	3.9.1
X	-	X	-	129. Emergency stopping system not as specified.	3.9.2
X	-	X	-	130. Parking system not as specified.	3.9.3
X	-	X	-	131. Air supply coupling not as specified.	3.9.4
X	-	X	X	132. Tires not as specified.	3.10
X	-	X	-	133. Wheels not as specified.	3.10.1
X	-	X	-	134. Valve parts not as specified.	3.10.2
X	-	X	X	135. Tire inflator not as specified.	3.10.3
X	-	X	-	136. Hoist system not as specified.	3.11
X	-	X	-	137. Reservoir not as specified.	3.11.1

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

TABLE VI. Examination Schedule (Cont'd)

1	2	3	4	5	6
X	-	X	-	138. Steering system not as specified.	3.12
X	-	X	-	139. Electrical system not as as specified.	3.13
X	-	X	-	140. Wiring not as specified.	3.13.1
X	-	X	X	141. Slave receptacle not as specified.	3.13.2
X	-	X	X	142. Lighting not as specified.	3.13.3
X	-	X	X	143. Blackout lights not as specified.	3.13.4
X	-	X	X	144. Instrument panel lights not as specified.	3.13.5
X	-	X	X	145. Light switch not as specified.	3.13.6
X	-	X	X	146. Light guards not as specified.	3.13.7
X	-	X	X	147. Turn signals not as specified.	3.13.8
X	-	X	X	148. Reflectors not as specified.	3.13.9
X	-	X	X	149. Circuit breakers not as specified.	3.13.10
X	-	X	X	150. Batteries not as specified.	3.13.11
X	-	X	X	151. Battery enclosure not as specified.	3.13.11.1
X	-	X	X	152. Alternator not as specified.	3.13.12
X	-	X	X	153. Cab not as specified.	3.14
X	-	X	X	154. Seat not as specified.	3.14.1
X	-	X	X	155. Seatbelts not as specified.	3.14.1
X	-	X	X	156. Heater and defroster not as specified.	3.14.2
X	-	X	X	157. Windshield wipers and washers not as specified.	3.14.3
X	-	X	X	158. Boarding provisions not as specified.	3.14.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TABLE VI. Examination Schedule (Cont'd)

1	2	3	4	5	6
X	-	X	X	159. Instrument panel not as specified.	3.14.5
X	-	X	X	160. Instruments, indicators and switches not as specified.	3.14.5
X	-	X	-	161. Dump body not as specified.	3.15
X	-	X	X	162. Rock ejectors not as specified.	3.15
X	-	X	-	163. Body fabrication not as specified.	3.15.1
X	-	X	-	164. Canopy fabrication not as specified.	3.15.2
X	-	X	X	165. Toolbox not as specified.	3.16
X	-	X	X	166. Fuel, lubricants and coolants not as specified.	3.17
X	-	X	-	167. Lubrication fittings not as specified.	3.17.1
X	-	X	-	168. Lubricant reservoir enclosure not as specified.	3.17.2
X	X	X	X	169. Fungus and moisture resistance not as specified.	3.18, 3.18.1
X	-	X	X	170. Treatment and painting not as specified.	3.19
X	-	X	X	171. Slinging and tiedown provisions not as specified.	3.20.1, 3.20.2
X	-	X	X	172. Identification and other required marking not as specified.	3.21
X	-	X	X	173. Operating and safety instructions missing.	3.21.2
X	-	X	X	174. Stencil markings not as specified.	3.21.3

Source: https://assist.dla.mil -- Downloaded: 20260-16T16:04Z
Check the source to verify that this is the current version before use.

TABLE VI. Examination Schedule (Cont'd)

1	2	3	4	5	6
X	X	X	X	175. Workmanship not as specified.	3.22
X	-	X	X	176. Electromagnetic interference suppression components and network not identical with approved model.	3.13.13
X	X	X	X	177. Permanent deformation of any part or assembly.	Section 3
X	X	X	X	178. Mechanical failure or deterioration of insulation.	Section 3
X	X	X	X	179. Discoloration due to overheating.	Section 3
X	X	X	X	180. Parts or components missing.	Section 3
X	X	X	X	181. Assembly incomplete or incorrect.	Section 3
X	X	X	X	182. Cracked or broken parts or assemblies.	Section 3
X	X	X	X	183. Contact of lubrication, air, fuel, or hydraulic lines with moving parts.	Section 3
X	X	X	X	184. Leaks in coolant, fuel, oil, or air lines.	Section 3

4.6.2 Tests. Tests shall be conducted in accordance with Table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TABLE VII. Test Schedule

Preproduction	Initial Production	Quality Conformance	Test	Test Method	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	X	-	Drawbar pull.	1	4.6.2.2	3.5.7
X	X	-	Performance.	2	4.6.2.3	3.5.3
X	X	-	Rainwind.	3	4.6.2.4	3.5.6
X	-	-	Fording.	4	4.6.2.5	3.5.5
X	X	-	Service brake system.	5	4.6.2.6	3.9.1
X	X	-	Emergency stopping system.	6	4.6.2.7	3.9.2
X	X	-	Parking system	7	4.6.2.8	3.9.3
X	X	-	Drawbar and towing.	8	4.6.2.9	3.5.8
X	X	-	High temperature.	9	4.6.2.10	3.5.1, 3.8 3.8.1, 3.8.6
X	X	-	Low temperature.	10	4.6.2.11	3.5.1, 3.8.2.2
X	-	-	Storage.	11	4.6.2.12	3.5.2
X	X	X	Operational.	12	4.6.2.13	Section 3
X	X	-	Emergency steering.	13	4.6.2.14	3.12
X	X	-	Retarder and downshift inhibitor.	14	4.6.2.15	3.8.6, 3.8.7
X	X	-	Electromagnetic interference.	-	4.6.2.16	3.13.13
X	X	-	Slinging provisions.	15	4.6.2.17	3.20.1
X	X	-	Tiedown provisions.	16	4.6.2.18	3.20.2
X	X	-	Ease of Maintenance.	17	4.6.2.19	3.7
X	X	-	Convoy ability.	18	4.6.2.20	3.5.4, 3.8 3.8.1, 3.8.6
X	X	-	Stability.	19	4.6.2.21	3.5.9

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TABLE VII. Test Schedule (Cont'd)

1	2	3	4	5	6	7
X	X	-	Vehicle clearance circle.	20	4.6:2.22	Table I
X	X	-	Sound level.	21	4.6.2.23	3.6.4.1, 3.6.4.2
X	-	-	ROPS and FOPS.	-	4.6.2.24	3.6.3

4.6.2.1 Test conditions.

(a) Test apparatus shall be accurate within the limits specified in SAE J872.
(b) Test methods referenced in tests specified herein are those contained in Appendix A.
(c) All working lights shall be turned on and left on during all performance tests.
(d) Unless otherwise specified herein, sequence of tests is optional with the contractor.
(e) Only that maintenance established by the contractor and submitted as a Maintenance Schedule prior to the start of the tests shall be performed during the test.
(f) Throughout the preproduction testing, a tachograph shall be installed on the truck to obtain permanent records of test conditions. The tachograph shall be Argo Instruments Corp., Model TCO-14, or equal, which will record the following information on a single 24-hour recording chart:

(1) Continuous recording of engine speed in rpm.
(2) Operating and stopping periods.
(3) Continuous recording of miles traveled.

Dual drive adapters shall be utilized in the tachometer and speedometer drive cables to provide simultaneous operation throughout the test of the tachograph and the on-vehicle speedometer and tachometer. The tachograph shall have a locked cover so that a key is required to change the chart. The completed charts will be made available to the Government representative on a daily basis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(g) The truck shall be serviced with the oils, greases and fuel specified in Table IV and designated for use in the ambient temperatures at which the tests will be conducted.
(h) The truck shall be limbered as prescribed by the contractor.
(i) The hydraulic system relief pressure shall be set as specified in 3.10. Any change made to this setting shall constitute failure of all tests.

4.6.2.2 Drawbar pull. Test the truck in accordance with Test Method No. 1. Nonconformance to 3.5.7 shall constitute failure of this test.

4.6.2.3 Performance. Test the truck in accordance with Test Method No. 2. Nonconformance to 3.5.3, inability of the truck to perform the operation, the necessity of repeated mechanical adjustment or replacement of any parts other than those specified in the maintenance schedule in order to complete operation successfully, any malfunctioning, mechanical or electrical failure (except lamp bulbs or sealed beam lamps), breakage of any component or attachment, or oil temperatures above 250° F shall constitute failure of this test.

4.6.2.4 Rain-wind. Test the truck in accordance with Test Method No. 3. Nonconformance to 3.5.6 shall constitute failure of this test.

4.6.2.5 Fording. Test the truck in accordance with Test Method No. 4. Inability of the truck engine to start and stop, evidence of excessive rust, corrosion or deterioration within the brake assembly or nonconformance to 3.5.5 shall constitute failure of this test.

4.6.2.6 Service brake system. Test the truck in accordance with Test Method No. 5. Nonconformance to 3.9.1 or inability of the brakes to bring the truck to a complete stop safely shall constitute failure of this test.

4.6.2.7 Emergency stopping system. Test the truck in accordance with Test Method No. 6. Nonconformance to 3.9.2 or inability of the brakes to bring the truck to a complete stop safely shall constitute failure of this test.

4.6.2.8 Parking system. Test the truck in accordance with Test Method No. 7. Nonconformance to 3.9.3 shall constitute failure of this test.

4.6.2.9 Drawbar and towing. Test the truck in accordance with Test Method No. 8. Nonconformance to 3.5.8 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

4.6.2.10 High temperature. The truck shall be tested in accordance with Test Method No. 9. Nonconformance to 3.5.1, inability to start the engine within 5 minutes, inability to operate the engine under rated load conditions within 15 minutes without unnatural control manipulation, failure of controls and instruments to operate as specified, or oil or coolant temperatures above those specified in 3.8, 3.8.1, or 3.8.6 shall constitute failure of this test.

4.6.2.11 Low temperature. Test the truck in accordance with Test Method No. 10. Nonconformance to 3.5.1 or 3.8.2.2, inability to start the engine within 5 minutes including cranking time, inability to operate under load conditions within 15 minutes after starting without unnatural or continued manipulation of controls, failure of controls or instruments to operate, inability of the operator to manipulate any control or adjustable device because of the bulk of arctic clothing, or inability to maintain a clear windshield with the defroster shall constitute failure of this test.

4.6.2.12 Storage. Test the truck in accordance with Test Method No. 11. Nonconformance to 3.5.2 or evidence of any of the following shall constitute failure of this test:

(a) Drainage, leaks, or cracks as a result of physical hardening or set of critical materials, differential contraction of fitted parts resulting in seizure and stress in metallic plumbing lines, or crystallization and fatigue of metals, alloys or plastics.
(b) Cracking or discoloration of instrument or gage plastics or glass.
(c) Disintegration of thermal or electrical insulation.

4.6.2.13 Operational. Test the truck in accordance with Test Method No. 12. Evidence of any deficiency or need for adjustment or maintenance shall constitute failure of this test.

4.6.2.14 Emergency steering. Test the truck in accordance with Test Method No. 13. Nonconformance to 3.12, inability of the operator to maintain complete control of the truck, or inability of the truck to contact all markers shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

4.6.2.15 Retarder and downshift inhibitor. Test the truck in accordance with Test Method No. 14. Nonconformance to 3.8.6 or 3.8.7, inability of the retarder to reduce the speed of the truck, or inability to downshift will constitute failure of this test.

4.6.2.16 Electromagnetic interference. The truck shall be tested to determine conformance to 3.13.13. The contractor shall furnish the contracting officer the report of the test required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6.2.17 Slinging provision. Test the truck in accordance with Test Method No. 15. Nonconformance to 3.20.1, weld failure, or permanent deformation of any truck component or attachment shall constitute failure of this test.

4.6.2.18 Tiedown provisions. Test the truck in accordance with Test Method No. 16. Nonconformance to 3.20.2, weld failure, permanent deformation of any truck component or attachment, or damage to any tiedown cable shall constitute failure of this test.

4.6.2.19 Ease of maintenance. Test the truck in accordance with Test Method No. 17. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.20 Convoy ability. Test the truck in accordance with Test Method No. 18. Nonconformance to 3.5.4, inability of the truck to maintain the required minimum average speed for a period of 3 hours, or oil or coolant temperatures above those specified in 3.8, 3.8.1 or 3.8.6 shall constitute failure of this test.

4.6.2.21 Stability. Test the truck in accordance with Test Method No. 19. Nonconformance to 3.5.9, evidence of any component not receiving lubrication, malfunction of any assembly or system, or inability of the truck to attain a speed of not less than the high-gear travel speed of the truck following slope runs shall constitute failure of this test.

4.6.2.22 Vehicle clearance circle. Test the truck in accordance with Test Method No. 20. A vehicle clearance circle greater than that specified in Table I shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

4.6.2.23 Sound level. Test the truck in accordance with Test Method No. 21. Nonconformance to 3.6.4.1 and 3.6.4.2 shall constitute failure of this test.

4.6.2.24 Rollover Protective Structure (ROPS) and Falling Objects Protective Structure (FOPS).

4.6.2.24.1 Test procedure. When a commercial structure(s) is used, the certification data shall be furnished to the contracting officer (see 3.6.3). When required, the structure shall be tested as specified in SAE J1040 and SAE J231.

4.6.2.24.2 Failure criteria. One or more of the following shall constitute failure of this test:

(a) The contractor does not furnish the certification data specified, or nonconformance to SAE J1040 and SAE J231, as applicable.
(b) The structure(s) does not conform to low temperature requirements.
(c) The structure(s) interferes with maintainability, operational visibility, ingress into or egress from the truck, or protrudes into the "space envelope" required by SAE J154.
(d) Seatbelts do not conform to SAE J386.
(e) Failure of the contractor to mount the information specified in 3.6.3 on the structure(s).

4.6.2.25 Posttest disassembly and examination. Upon completion of all tests specified in 4.3.2, measure and record the compression in each engine cylinder and dismantle the truck sufficiently to permit detailed disassembly and visual examination of the following components:

(a) Engine-cylinder head, all valves, two pistons and piston pins from the two cylinders having the lowest compression readings at the end of the test, one main bearing and two connecting rod bearings, and the turbocharger, if used.
(b) Torque converter and transmission.
(c) Universal joints and drive shafts.
(d) Differentials from front and rear axle clutch components, if applicable.
(e) One final drive planetary.
(f) Complete brake system components.
(g) Steering system cylinders, valves and pump.
(h) Hydraulic system-pump valves, one cylinder of each size, control levers, and any hose and fitting where leakage occurred during the tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(i) One front and one rear wheel and wheel bearing.
(j) Structural parts - welds at high stress points.

Additional disassembly and examination shall be made at the option of the contracting officer. The disassembly and examination shall consist of a paved shop area of sufficient size to permit the disassembly and detailed examination of all truck components. The area should be well lighted, clean, and have provisions for prohibiting all but required personnel. Available equipment shall include an engine test stand, tables and benches to permit parts grouping, flashlights, and photographic equipment. (The truck shall be examined in accordance with Table VI, Column 2.) Record complete description of the condition of all parts and assemblies examined, and take pictures of any deficiency. Each evidence of corrosion which affects the functions of the truck, each evidence of permanent deformation, breakage, or excessive wear of any component shall constitute a defect. Manufacturing tolerances listed in the contractor's quality control program shall be made available upon request and shall be used as the basis for determining excessive wear.

4.7 Inspection comparison. The Government may select trucks at any time during the contract production period and subject these trucks to the examination specified in 4.6.1 and Column 3 of Table VI and to the tests specified in Column 1 of Table VII to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on trucks selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production trucks. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison truck and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on trucks

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all trucks accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging. The preservation, packaging, packing, and marking of the truck shall be examined to determine compliance with the quality assurance provisions of MIL-STD-281 and the exceptions and additional requirements specified in Section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Each complete truck shall be preserved and packaged in accordance with MIL-STD-281, Level A (mobile) with the following exceptions and additional requirements.

5.1.1.1 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, Level A, Type I, Method I.

5.1.1.2 Batteries. Batteries shall be secured in place in the battery carrier as specified in MIL-STD-281.

5.1.1.3 Cooling system. The cooling system shall be preserved by the water and antifreeze procedure as specified in MIL-STD-281.

5.1.1.4 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.1.1.5 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40, Level A.

5.1.1.6 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

5.1.2 Level B. Each complete truck shall be preserved and packaged in accordance with MIL-STD-281, Level B (mobile) and with the exceptions and additional requirements specified in 5.1.1 for Level A.

5.1.3 Level C. Each complete truck shall be preserved and packaged in accordance with MIL-STD-281, Level C (mobile) and the cooling system preserved by the water and antifreeze procedure specified herein.

5.2 Packing. Each complete truck, preserved and packaged as specified in 5.1 shall not require packing.

5.3 Marking. Packages and mobile trucks shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The trucks covered by this specification are intended primarily for rock and earthmoving construction work where distances are too great for economical use of crawler or rubber-tired earthmoving equipment, and the speed, maneuverability and capacity of a heavy-duty, off-highway dump truck is required. The truck is suitable for use in military operations such as:

(a) Operating at altitudes between sea level and a 5000-foot elevation.
(b) Emergency towing by another vehicle.
(c) Operating in and outside storage under weather conditions specified herein.
(d) Impact forces encountered in rail shipments.
(e) Withstanding convoy operations.
(f) Hauling stone and other material over nonmaintained roads with slopes up to 15 percent.
(g) With empty dump body negotiating uphill and downhill grades from 0 to 30 percent and sidehill slopes from 0 to 17.5 percent.
(h) Fording 30-inch-deep freshwater streams.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of truck required (see 1.2).
(c) Time frame required for submission of preproduction model, and number of trucks required (see 3.2).
(d) When the Government will conduct any or all of the preproduction examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(e) When initial production inspection is required and number of trucks to be furnished (see 3.3).
(f) When commercial standard couplings are required (see 3.9.4).
(g) When a tire inflator is required (see 3.10.3).
(h) When commercial standard electrical system is required (see 3.13).
(i) When lights other than those specified are required (see 3.13.3).
(j) When a heater and defroster are not required (see 3.14.2).
(k) Applicable ambient temperature range governing weights of lubricants with which the truck is to be serviced in preparation for shipment (see 3.17.3).
(l) Color required (see 3.19).
(m) Level of preservation and packaging required (see 5.1).
(n) Time frame required for submission of Technical Information Package (see 3.23).

6.3 Preproduction model. Any changes or deviations of production trucks from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his oblication to furnish trucks conforming to this specification.

Custodians:
Army - ME
Navy - YD

Review activity:
Army - AT

User activity:
Army - CE

Preparing activity:
Army - ME

Project 3805-0107

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

APPENDIX

FOR

TRUCKS, DUMP, OFF-ROAD, DIESEL-ENGINE-DRIVEN,

15 TO 50 TON PAYLOAD

10. SCOPE. The test methods contained within this appendix are to determine whether dump trucks procured under this specification conform to the requirements set forth.

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 1

DRAWBAR PULL

1. TEST COURSE:

(a) The test course shall be a dry concrete surface with not more than 1 percent average longitudinal grade. The crown slope shall not exceed 1.5 percent, measured from center to outer edge. The width of the straight portions shall be wider than the vehicle and long enough to meet the requirements of the test method.

2. TEST APPARATUS:

(a) General apparatus: (Alternate approved equipment may be used.)

(1) Barometer.
(2) Gage, tire pressure.
(3) Psychrometer.
(4) Load cell, hydraulic or electronic, w/amplifier and recorder (use lockout device, when possible, to prevent overload of instruments during turns and accelerations).
(5) Odometer.
(6) Recorder, totalizing.
(7) Scale, platform.
(8) Tachometer.
(9) Tape, measuring, surveyor's, chain.
(10) Thermometer.
(11) Stopwatch.
(12) Electronic recorder to obtain speed and elapsed time.
(13) Robert Bosch sampling pump - EFAW 65.
(14) Robert Bosch evaluating instruments EFAW 68.

(b) Special apparatus:

A towed load is required with a drag force adjustable as necessary to control:

(1) The speed of towing vehicle output shaft constant within plus or minus 3 percent of a selected speed, or

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(2) the reserve tractive force constant within plus or minus 2 percent of the selected load.

3. TEST PROCEDURE:

(a) Test shall be conducted with the rated load in the dump body.
(b) Determine the distance the truck travels per revolution of the drive wheels without tractive load. Count revolutions of drive wheels and divide distance traveled by revolutions to obtain travel per revolution. The following methods of determining distance of travel per revolutions are acceptable:

(1) Drive the truck in lowest gear with engine throttled to its lowest uniform running speed and without directional correction for a distance of not less than 100 feet.
(2) When truck drag does not exceed 100 pounds per ton, tow or push the truck at the lowest practical uniform speed to determine the distance traveled per revolution of the drive wheels.

(c) Attach load to be towed to the drawbar.
(d) Record general data required by test log (Figure A-2).
(e) Record, while truck and load are traveling over the test course with towed load adjusted to maintain the output shaft speed at specified rpm, and with the towing cable level:

(1) Drawbar pull.
(2) Time.
(3) Distance.
(4) Number of revolutions of engine and of output shaft.
(5) Number of revolutions of each rear wheel.

(f) Recorded test runs shall cover a minimum of 0.5 minute or a minimum distance of 100 feet, whichever is greater. Use the average of four runs (two in each direction to report truck performance at each selected speed or load).
(g) Travel during the recorded runs shall not exceed 1-1/2 feet to right or left of an established centerline. Revolutions of the drive wheels shall not vary from each other more than 3 percent. Use a counter on each rear wheel to register revolutions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(h) During any recorded run, the instantaneous speed of the output shaft shall not vary more than plus or minus 3 percent from the specified speed. The average output shaft speed for any one run shall not vary more than plus or minus 2 percent from the specified speed and the average for the four selected runs shall not vary more than plus or minus 1 percent from the specified speed.

(i) Alternate procedure:

(1) Control reserve tractive force and hold constant within plus or minus 2 percent for each run.

(2) Record data as required in (d) above. The instantaneous engine and vehicle ground-speed values at the end of the test run shall be within 3 percent of those at the start.

(j) Calculations:

$$\text{Travel efficiency, percent} = \frac{S_o \times 100}{S_s}$$

$$\text{Vehicle speed, mph} = \frac{S}{88\ t}$$

$$\text{Reserve tractive horsepower } (thp_o) = \frac{F_o \times \text{mph}}{375}$$

$$\text{Average speed of engine, rpm } (N_e) = \frac{N_e}{t}$$

$$\text{Average speed of output shaft, rmp } (N_o) = \frac{N_o}{t}$$

Where:

F_o = reserve tractive force, pounds averaged for either time or distance
t = time to travel test distance, minutes
N_e = revolutions of engine to travel test distance
N_o = revolutions of output shaft to travel test distance

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

N_w = average revolutions of drive wheels to travel distance
S^w = Test distance traveled, feet
S_s = average distance vehicle traveled without tractive load per revolution of the drive wheels, feet
S_o = average distance vehicle traveled with tractive load per revolution of the drive wheels

$$\text{feet} = \frac{S}{N_w}$$

Note. Engine corrections for atmospheric conditions, other than altitude, shall not be used.

(k) Repeat the above procedures to include the approximate ground speeds in each gear as follows:

(1) 20 percent of maximum ground speed.
(2) 40 percent of maximum ground speed.
(3) 60 percent of maximum ground speed.
(4) 80 percent of maximum ground speed.

(l) Using the data obtained, plot drawbar pull versus speed for each gear ratio. Draw a curve through the intersect points of the successive gear-ratio curves and compare with the appropriate curve shown in Figure A-1.

(m) Maintain the truck in fully serviced condition throughout the test. Inflate truck tires in accordance with truck manufacturer's recommendations. Take smoke density measurements at or near maximum power point in first and second gear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

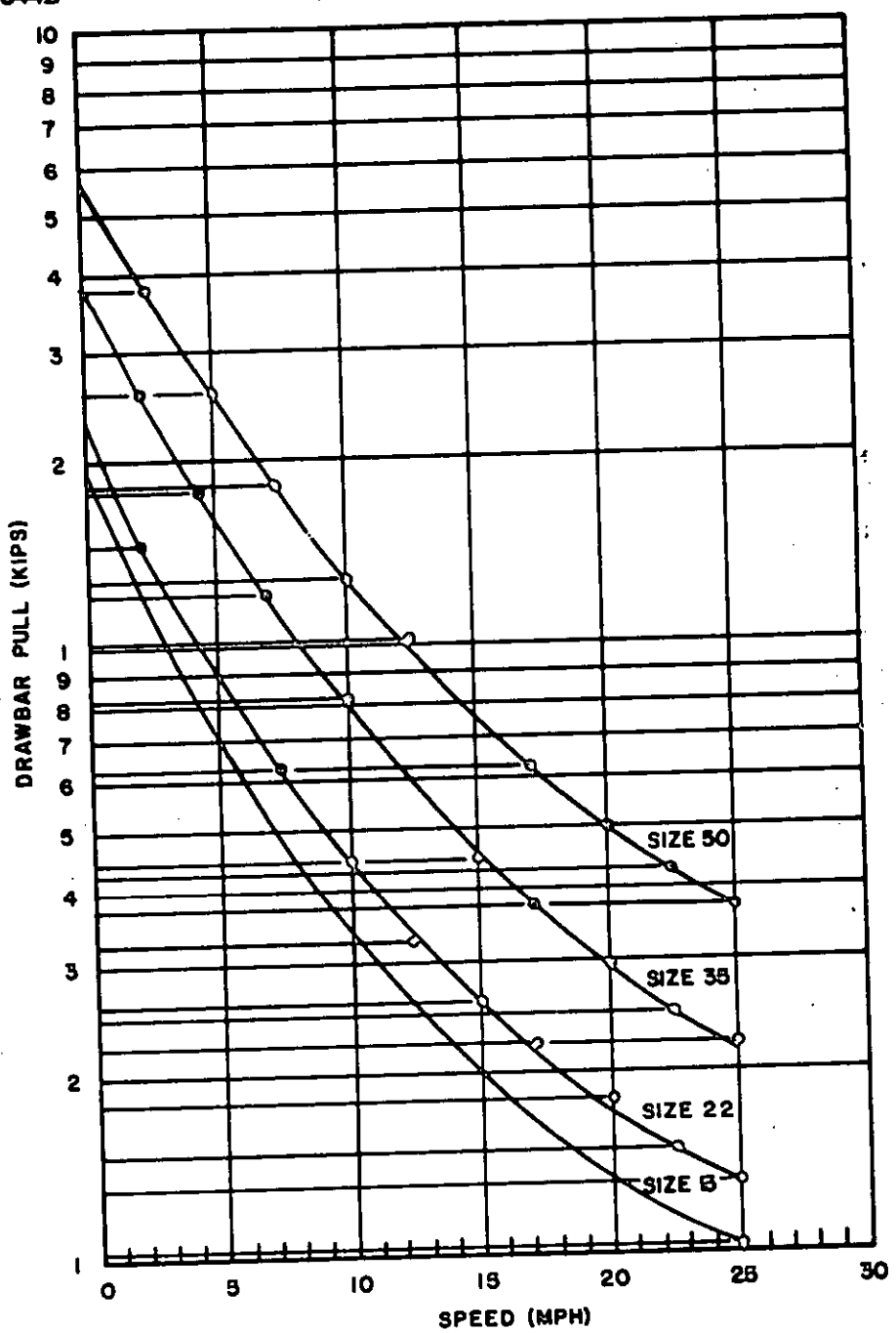


FIGURE A-1. Drawbar pull versus travel speed. X-3228

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

RESERVE TRACTIVE ABILITY TEST SUMMARY

GENERAL DATA

VEHICLE: MAKE _____ MODEL _____ SERIAL NO. _____
VEHICLE ATTACHMENT: TYPE _____
MAKE _____ MODEL _____ SERIAL NO. _____
ENGINE: MAKE _____ MODEL _____ SERIAL NO. _____
RATED RPM _____ FUEL _____
TEST COURSE: MATERIAL _____ CONDITION _____
LOCATION _____

AXLE: LOAD _____ FRONT: _____ REAR: _____
TIRE: TYPE _____ SIZE _____
INFLATION PRES: _____
GROSS VEHICLE WEIGHT _____ LB
HITCH HEIGHT ABOVE GROUND: _____ IN.

AVERAGE DATA ON INDIVIDUAL TESTS

TEST NUMBER .		1	2	3	4	
LOG SHEET NO.						
TRANSMISSION GEAR						
OVER-ALL RATIO						
ENGINE SPEED (RPM)	N_E					
OUTPUT SHAFT (RPM)	N_O					
GROUND SPEED	MPH					
RESERVE TRACTIVE FORCE	F_O					
TRAVEL EFFICIENCY *	θ					
RESERVE TRACTIVE HP	THP_O					
AMBIENT TEMPERATURE	F					
BAROMETER	IN HG					
RELATIVE HUMIDITY	%					

* STATE WHETHER VEHICLE WAS DRIVEN ( ), TOWED ( ), OR PUSHED ( ) TO OBTAIN DISTANCES TRAVELED PER REVOLUTION OF DRIVE WHEEL OR SPROCKET.

DATE

FIGURE A-2. TEST LOG FOR DRAWBAR PULL TEST

X-2471

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 2

PERFORMANCE

1. TEST COURSE:

(a) The test course shall be an unpaved, secondary, construction or quarry haul road. At least 1/2 mile of the course shall be grades of not less than 15 percent. The overall length of the test course shall be not greater than 1-1/2 miles. A commercial operation is acceptable provided prior approval is obtained from the contracting officer.

2. TEST APPARATUS:

(a) Material for loading truck.
(b) Gage, tire pressure.
(c) Odometer, calibrated.
(d) Stopwatch.
(e) Instrument for measuring grade of road.
(f) Loader compatible with size of truck.
(g) Scales for weighing truck.

3. TEST PROCEDURE:

(a) Arrange sufficient loading material at one end of the test course for maintaining truck cycle operations for a minimum of 1 hour but no more than 4 hours. Load the truck to no less than rated GVW with the loader. Drive the loaded truck on the test course. At the end of the course, dump the load and return with the dump body empty. Continue the cycle for the time period selected above (1 to 4 hours). At the end of the time period, reverse the cycle by hauling material in the opposite direction for the same time period. Continue operation for a minimum of 500 engine hours of production operation. Production hours shall not include waiting time for the loader to load another truck, nor times exceeding 10 minutes for loading the test vehicle. The loaded truck should be weighed and the tire pressure checked at least once per day. Periodic maintenance of the haul road is acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(b) Observe and record:

(1) Length and grades of test course.
(2) Direction of each cycle.
(3) Time required for each cycle including time for each direction.

(c) The truck will be modified in accordance with MIL-STD-1400, Test Method No. 1410 for a minimum of 150 hours of the performance test. Upon completion of this period, the altitude restriction shall be removed and the engine returned to the intermittent horsepower rating at the specified rpm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 3

RAIN-WIND

1. TEST APPARATUS:

   (a) Timing device.
   (b) Metering device.
   (c) Spray fixture.
   (d) Wind generator and wind velocity gage.
   (e) A means to trap and collect water which may enter the exhaust system leading to the exhaust manifold.

2. TEST PROCEDURE:

   (a) Park the truck on a horizontal surface and expose to natural or simulated rain falling toward the truck at an angle of approximately 45 degrees to the horizontal. The volume of rain impinging on the horizontal surface shall be not less than 1/4 gallon per minute per square foot of wetted, horizontal area (24 inches of rain per hour). Simulated rain shall originate from a source which produces droplets of uniform distribution and velocity similar to natural rain at the point of contact with the truck. The rain shall then be driven horizontally onto and against the center of mass of the truck by a generated wind of not less than 30 miles per hour. If equipped with side panels, the panels shall be removed for this test. The hood or top panel shall not be removed.
   (b) Storage condition:

      (1) Position the truck to expose the exhaust opening to the most critical direction for water infiltration with respect to the wind. Shut down engine.
      (2) Insert item 1(e) in the exhaust system at a location where it will trap and collect water entering the exhaust system.
      (3) Expose the truck in this position to the rain as described in 2(a) for a period of 5 minutes.
      (4) Stop the wind generator and spray rain fixture.
      (5) Examine the water-collecting means for presence of water.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

(c) Operating.

(1) Expose the top and left side to the rain as described above for a period of 5 minutes with the engine shut down.
(2) With water flowing, start engine within 2 minutes (without removing accumulated water) and allow engine to idle for 2 minutes.
(3) Repeat (1) and (2) with the top and front exposed.
(4) Repeat (1) and (2) with the top and right side exposed.
(5) Repeat (1) and (2) with the top and rear exposed.

(d) At the conclusion of the test, examine the truck for the following and record observations:

(1) Water in the air-intake filter.
(2) Water leakage into the engine crankcase sump, exhaust system and hydraulic reservoir.
(3) Performance of the electrical components, gages and instruments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 4

FORDING

1. TEST COURSE:

(a) The test course shall be a body of freshwater not less than 30 inches deep and of sufficient size to permit the truck to enter.

2. TEST APPARATUS:

None.

3. TEST PROCEDURE:

(a) Prior to test, take oil samples from the engine and all hydraulic and lubricant reservoirs, and analyze for water content.
(b) Run the truck engine until the engine coolant reaches operating temperature as indicted by the temperature gage on the truck before the truck enters the water. Operate the truck in first gear forward and reverse a minimum of 15 minutes in water at a depth of 30 inches as indicated by a line marked on the truck. Then operate out of water in high gear for 15 minutes.
(c) Perform the above test five times, except that during the second, third, fourth, and fifth tests, the truck engine shall be started and stopped a minimum of two times while in the water at a 30-inch depth.
(d) Upon completion of the water fording teats, park the truck on a dry surface, shut off the engine, and let the truck stand for a period of 1 hour. Then take oil samples from the engine and all hydraulic lubricant reservoirs and analyze for water content.
(e) Remove one wheel and examine wheel brake assembly for evidence of rust, corrosion or deterioration.

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 5

SERVICE BRAKE

1. TEST COURSE:

(a) The test course shall be dry, level, concrete pavement of sufficient length to attain the required speed and safely stop the truck.

2. TEST APPARATUS:

(a) Brake-pedal-actuated marking device.
(b) Fifth wheel attachment.
(c) Timer.
(d) Measuring tape.

3. TEST PROCEDURE:

(a) Mount the automatic marking device on the truck so that it actuates within the first 1/4-inch of travel of the brake pedal, to locate the point of brake application on the concrete surface. Drive the truck at rated GVW over the course at a speed of 40 mph. Apply the service brakes. Measure and record the distance between the point of brake application and the point where the truck stops. Repeat the procedure again, 24 times in succession, allowing not more than 3 minutes between each braking stop for brake cooling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 6

EMERGENCY STOPPING STSTEM

1. TEST COURSE:

(a) The test course shall be dry, level, concrete pavement of sufficient length to attain the required speed and safely stop the truck.

2. TEST APPARATUS:

(a) Brake-actuated marking device.
(b) Fifth-wheel attachment.
(c) Timer.
(d) Measuring tape.

3. TEST PROCEDURE:

(a) Mount the automatic marking device on the truck so that it actuates when the emergency stopping system is applied to locate the point of brake application on the concrete surface. Drive the truck at rated GVW over the course at a speed of 20 mph. Apply the emergency stopping system by simulating the most critical single failure in the service brake system. Measure and record the distance between the point of brake application and the point where the truck stops. Repeat 3 more times.

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 7

PARKING SYSTEM

1. TEST COURSE:

(a) The test course shall be dry, swept, concrete surface with a minimum longitudinal grade of 30 percent with a minimum length equal to three times the overall truck length.

2. TEST PROCEDURE:

(a) Drive the truck at rated GVW onto the test course with the front of the truck in the downhill direction and stop. Apply the parking brake and dismount.
(b) Repeat the test with the truck positioned with the front in the uphill direction.

3. ALTERNATE TEST PROCEDURE:

(a) Drive the truck at rated GVW onto a dry, swept, concrete surface with a grade not exceeding 1 percent. Apply the parking brake with a force not exceeding 90 pounds. Apply a pull equal to 29 percent of the GVW of the truck at the rear drawbar in a direction parallel to the longitudinal axis of the truck.
(b) Repeat the test with the pull applied at the front drawbar.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 8

DRAWBAR AND TOWING

1. TEST COURSE:

(a) The test course shall be an improved and paved roadway surface having a length of continuity to permit speeds required.

2. TEST APPARATUS:

(a) Load cell (or equivalent approved means of determining applied load).
(b) Restraint (either tiedown eyes embedded in concrete or restraint vehicle).
(c) Tow vehicle with either connecting towbar or cable.

3. TEST PROCEDURE:

(a) Drawbar.

(1) With the truck restrained, apply a horizontal load on the front drawbar along the longitudinal axis of the truck equal to two times the applicable maximum drawbar pull requirement shown in Figure A-1.
(2) Maintain the applied load and examine the drawbar.
(3) Release the load and examine the drawbar.
(4) Repeat (1) thru (3) for the rear drawbar.

(b) Towing.

(1) Affix the truck at rated GVW to the towing vehicle by means of a towbar or cable.
(2) Tow the truck for a distance of 5 miles at an average speed of 10 mph. Where towing could damage the transmission, disconnect the power train.
(3) After towing, examine the drawbar.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 9

HIGH TEMPERATURE

1. TEST CONDITIONS:

(a) The truck, or truck assembly without dump body and rear axle assembly, shall be subjected to an ambient temperature of not less than plus 125° F until stabilized at the ambient temperature. The truck or truck assembly shall be direct-coupled to a dynamometer for this test. A chassis dynamometer is acceptable.

2. TEST APPARATUS:

(a) Test chamber.
(b) Thermocouples.
(c) Recording potentiometers.
(d) Thermometers (wet and dry).
(e) Dynamometer with all necessary instrumentation.
(f) Tachometer.
(g) Barometer.

3. TEST PROCEDURE:

(a) Install thermocouples as listed below:

(1) Engine coolant temperature: One in the engine-cooling-jacket discharge pipe to radiator top tank and another in the coolant inlet to the engine.
(2) Engine lubricating oil: In the oil gallery and oil sump.
(3) Fuel temperature: In the tank.
(4) Battery temperature: In the two center cells of each battery between the plates and at mid-depth of the electrolyte.
(5) Ambient temperature measured at a horizontal angle of 45 degrees from the sides and not less than 3 feet from each corner of the truck at engine-crankshaft level.
(6) Torque converter discharge line (to cooler).
(7) Torque converter sump.
(8) In the hydraulic reservoir near the pump inlet.
(9) Transfer case sump.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(b) Start the engine and run for 10 minutes, then record at 10-minute intervals temperature reading of the thermocouples listed in (a) during the operations specified in 3(c).

(c) Apply for 1 hour a load equal to 100 percent of the net maximum power rating of the engine (for this application) at the governed speed. At the end of the 1-hour period, change the load and cycle as follows for 30 complete cycles:

(1) Maximum rated load (as detetmined above) - 3 minutes. During every other cycle, use the next highest or lowest gear to obtain maximum rated load. For instance, if third gear is used during first cycle, use second or fourth gear for next cycle.
(2) 1/2 maximum rated load - 2 minutes.
(3) Zero load - 1 minute.

(d) Record the following readings at 10-minute intervals during the 4 hour operation:

(1) Revolutions per minute.
(2) Dynamometer torque.
(Take readings (1) and (2) simultaneously)
(3) Barometer.
(4) Wet- and dry-bulb air temperatures.

(e) During operation in (c) above, operate, at least once, all controls and switches other than starting switch and ether prime control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 10

LOW TEMPERATURE

1. TEST FACILITY

(a) The test facility shall be a test chamber of sufficient size and capacity to contain the truck, with or without dump body, and to provide temperatures no higher than minus 25° F.

2. EQUIPMENT:

(a) Temperature-controlled chamber, thermocouples and appropriate instrumentation, timing device, and appropriate cold-weather clothing.

3. PROCEDURE:

(a) Service the truck with antifreeze, lubricants, fuel, and hydraulic oil as specified in Table V. Install thermocouples at all points required to read the temperatures specified. The chamber exhaust system shall remove all engine exhaust fumes from the chamber during the test. Subject the truck to a temperature no higher than minus 25° F until stabilized; then start the engine. Record the temperature continuously or at intervals not greater than 5 minutes. Crank the engine by the batteries which are on the truck and which have been stabilized. Substitute batteries or other external cranking energy shall not be used. Operate the engine for 15 minutes. Operate the truck controls for 30 minutes utilizing the following procedure:

Cycle engine every 4 minutes for 1 minute duration from medium idle to to engine governed speed. Continue the cycle for 1/2 hour. Once every 15 minutes, apply the parking brake, block the wheels, shift to highest gear, and operate the engine at wide-open-throttle for 30 seconds.

Record temperatures at the following locations:

(1) Engine coolant inlet to the engine and discharge to radiator top tank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(2) Engine lubricating oil temperature in the oil gallery and oil sump.
(3) Torque converter oil temperature at the outlet from the converter to the cooler.
(4) Hydraulic oil temperature at the midpoint of the reservoir.
(5) Ambient temperature at a horizontal angle of 45 degrees from the sides and ends of the truck and not more than 3 feet from each corner of the truck at engine-crankshaft level.

The operator shall wear arctic clothing, including gloves and boots, during the test. Observe heater and defroster operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 11

STORAGE

1. TEST FACILITY:

   (a) The test facility shall be a test chamber of sufficient size and capacity to contain the truck, with or without dump body, and to provide temperatures ranging from minus 70° F to plus 155° F.

2. TEST APPARATUS:

   (a) Thermocouples.
   (b) Recording potentiometer.

3. TEST PROCEDURE:

   (a) Install thermocouples at all points required to record the temperature specified in Test Method No. 9.
   (b) Low-temperature storage.

      (1) Subject the truck to a low ambient temperature until the temperature of all parts is stabilized at not higher than minus 70° F. Maintain the temperature of all parts for a period of 12 hours.
      (2) Return the truck to ambient temperature, start the engine, and operate all controls.
      (3) Examine the truck.

   (c) High temperature storage.

      (1) Subject the truck to a high ambient temperature until the temperature of all parts is stabilized at not lower than plus 155° F. Maintain the temperature of all parts for a period of 4 hours.
      (2) Return the truck to ambient temperature, start the engine, and operate all controls.
      (3) Examine the truck.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the currentversion before use.

TEST METHOD NO. 12

OPERATIONAL

1. TEST COURSE:

   (a) The test course shall be an unimproved (not hard surfaced) road.

2. TEST MATERIALS:

   (a) A load equal to or greater than the rated capacity of the truck.

3. TEST PROCEDURE:

   (a) Operate the truck for 2 hours in all gears forward and reverse over the test course with the test load in the dump body. Truck operation shall include, but shall not be limited to, operation at maximum speed in all gear ranges, maneuvering through maximum right and left turns, and brake applications at both low and high speed. Include not less than three starts in first and second gear. Stop and restart the engine no less than three times.
   (b) With the test load in the dump body, partially raise and lower the body, without dumping the load, for 1 hour. With the body empty, cycle the hoist from maximum retraction to maximum extension for 1 hour.
   (c) During the foregoing operation, check all lights, switches and controls for proper operation. Paint damage during this test shall be spot painted after the test. Surfaces spot painted shall not constitute failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 13

EMERGENCY STEERING

1. TEST COURSE:

(a) The test course shall be a dry, level surface of sufficient length for the truck to attain a speed of not less than 25 mph. The course shall also include a curve with a maximum centerline radius of 100 feet in accordance with Figure A-3.

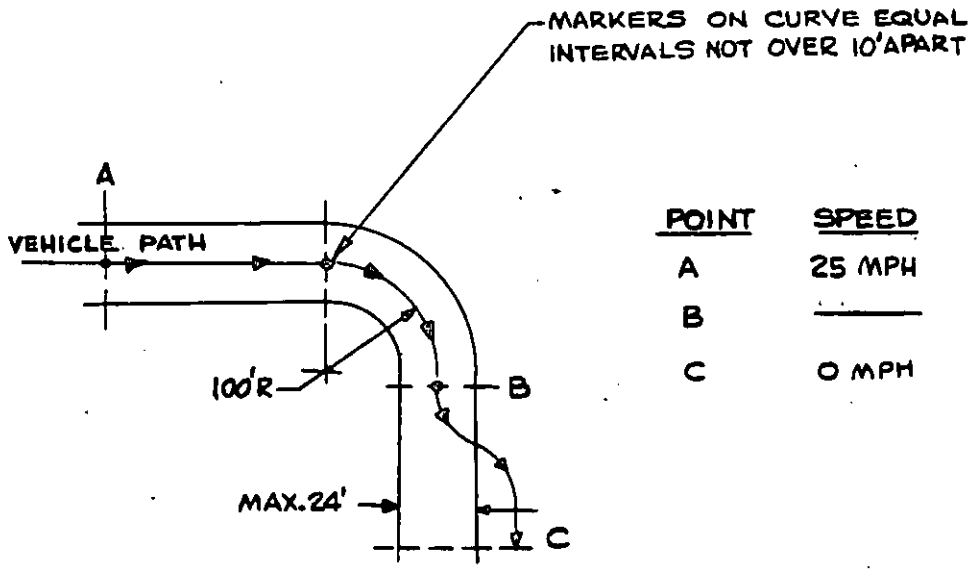
2. TEST APPARATUS:

(a) Tape, measuring (100-ft.).
(b) Markers (20 each of sufficient length to extend 36 inches above ground level) equally spaced along curve centerline.
(c) Speed indicator.
(d) Material equal to the payload and body capacity of the truck.

3. TEST PROCEDURE:

(a) Load the truck to rated GVW. Drive the truck a sufficient distance to attain a speed of not less than 25 mph. When the truck reaches Point A (Figure A-3) traveling 25 mph, shut off the engine and shift the transmission into neutral. Slow the truck by use of the service brakes if necessary. At Point B, steer the truck to the outside of the road away from the curve and pull off onto the shoulder of the road. Stop at Point C. The driver should maintain complete control of the truck at all times during the test. All markers on the 100-foot radius curve should be contacted by the truck while negotiating the curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.



X-2598

FIGURE A-3. TEST COURSE FOR EMERGENCY STEERING TEST.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 14

RETARDER AND DOWNSHIFT INHIBITOR

1. TEST COURSE:

(a) The test course shall be an improved surface area with a level section and a longitudinal grade. The level section should be sufficiently long enough to allow the truck to attain 15 mph. At one end of the level section, a descending slope shall have a minimum longitudinal grade of 30 percent with a minimum length equal to five times the overall truck length.

2. TEST APPARATUS:

(a) Material equal to payload and body capacity of the truck.
(b) Odometer.

3. TEST PROCEDURE:

(a) Calibrate the odometer to be used in the test.
(b) Start at the end of the level section of test course and accelerate the truck with rated GVW up to a minimum of 15 mph. At the crest of the incline, actuate the retarder and observe truck speed down the incline. Brakes other than the retarder shall not be used down the incline. Record truck speed at bottom of incline.
(c) Repeat 3(b), but instead of actuating the retarder, attempt to downshift the truck.
(d) Repeat 3(b), but instead of actuating the retarder, allow engine to return to low idle.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 15

SLINGING PROVISIONS

1. TEST APPARATUS:

(a) Crane or lifting device.
(b) Slings with hooks.

2. TEST PROCEDURE:

(a) Lift the truck by the lifting attachments and hold in suspension for at least 3 minutes in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the truck. Spreader bars are permitted. Determine the force and the angles of application for each attachment. Measure the clearance between each sling and the truck.
(b) Restrain the truck by anchoring the main frame or axles and subject each attachment to a load applied in the direction as determined in (a) and equal to 2-1/2 times the force determined in (a). Hold each load a minimum of 90 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 16

TIEDOWN PROVISIONS (DYNAMIC)

1. TEST APPARATUS:

(a) Timing devices.
(b) Tiedown material - cable not less than 1/2-inch size.
(c) Six railroad flatcars.
(d) Means of applying and holding constant load.
(e) Load measuring device.

2. TEST PROCEDURE:

(a) Anchor the truck on a flatcar using only the tiedown provisions. Use no blocking. Hump the car, with the truck tied down, at a speed of not less than 8 mph against five cars with their brakes set. The truck shall face in the direction of the five cars. No cushioning devices on the tiedowns or on the flatcars shall be used.
(b) Repeat the procedure with the rear of the truck in the direction of the five cars.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 17

EASE OF MAINTENANCE

1. TEST FACILITY:

(a) The test facility shall consist of an improved or paved surface or shop area of sufficient size to permit unobstructed surveillance of maintenance.

2. TEST APPARATUS:

(a) Common handtools.
(b) Special tools (if furnished by the contractor with the trucks).

3. TEST PROCEDURE:

(a) Examine the truck for accessibility of major assemblies.
(b) Examine position of drains with respect to accessibility.
(c) Using the tools, demonstrate that major assemblies are accessible for repair and maintenance.
(d) Demonstrate the accessibility of drains and path of discharge of lubricants and coolants by activating drainage controls.
(e) Perform and time each operation on the maintenance operation list (see 3.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 18

CONVOY ABILITY

1. TEST COURSE:

(a) The test course shall be an improved or paved road surface having a length or continuity to demonstrate speeds required.

2. TEST APPARATUS:

(a) Odometer.
(b) Stopwatch.
(c) Thermocouples and appropriate instrumentation.

3. TEST PROCEDURE:

(a) Calibrate the odometer to be used in the test by observing the distance recorded by the odometer in a series of three passes over a measured distance not less than 1 mile in length.
(b) To measure temperatures, install thermocouples as follows:

(1) In the engine oil gallery.
(2) In the engine oil discharge line to the cooler.
(3) In the engine sump.
(4) In the converter oil discharge line to the cooler.
(5) In the converter sump.
(6) In the transmission sump.
(7) In the engine coolant discharge line.

(c) Drive the truck with 1/2 rated load in body over the test course for a period of 3 hours.
(d) Record the total distance covered and the elapsed time.
(e) Compute the average speed over the 3-hour period.
(f) During the 3-hour run, record continuously, or at intervals not greater than 15 minutes, the temperatures measured at the locations described in 3(b).

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 19

STABILITY

1. TEST COURSE:

   (a) Slopes of not less than 17.5 percent and 30 percent compacted and graded. The width of the 17.5 percent slope and the length of the 30 percent slope shall be not less than 40 feet plus the length of the truck.
   (b) Straight, level surface of sufficient length to permit the truck to attain convoy speed.

2. TEST PROCEDURE:

   (a) Adjust the engine oil level to the low mark on the engine oil disptick. Drain or add fuel to begin the test with 1/4 tank of fuel.
   (b) Operate the truck engine for 1 hour on side slopes of 17.5 percent and longitudinal slopes of 30 percent. A holding cable may be used for safety purposes. Operations shall consist of starting, running, and stopping the engine, both the right side and left side of the truck on the downgrade of the slopes, and both ascending and descending the longitudinal slopes for approximately 15 minutes in each direction. Turning while on side slopes is not required. At the end of 1 hour, operate the truck for 15 minutes on level surface at convoy speed.
   (c) Observe and record:

      (1) Oil pressure readings at 5-minute intervals.
      (2) Engine coolant temperature readings at 5-minute intervals.
      (3) Speed attained after slope operation.
      (4) Ambient temperature during test.
      (5) Transmission sump temperature readings at 5-minute intervals.

Source: https://assist.dla.mil -- Downloaded: 202610-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 20

VEHICLE CLEARANCE CIRCLE

1. TEST COURSE:

(a) The test course shall be a dry, level, concrete surface of sufficient area.

2. TEST APPARATUS:

(a) Plumb bob.
(b) Measuring tape.
(c) Chalk.

3. TEST PROCEDURE:

(a) Measure the diameter of the smallest circle which will enclose the outermost points of projection of the vehicle while executing its sharpest practicable turn.

Source: https://assist.dla.mil -- Downloaded: 202640-16T16:04Z
Check the source to verify that this is the current version before use.

TEST METHOD NO. 21

SOUND LEVEL

1. TEST COURSE:

(a) The test course shall be in accordance with SAE J919 and SAE J88.

2. TEST APPARATUS:

(a) In accordance with SAE J919 and SAE J88.

3. TEST PROCEDURE:

(a) Measure the sound level at the operator's station in accordance with SAE J919 under the following conditions:

STATIONARY

(1) Maximum attainable governed engine speed at no-load.
(2) Maximum attainable governed engine speed at torque converter stall.
(3) Maximum attainable governed engine speed while dumping rated load.

MOVING

(4) Full governed control setting in the intermediate forward gear loaded with any combination of rolling resistance, drag load, or vehicle brake to obtain rated engine speed.

(b) Measure the exterior sound level in accordance with SAE J88.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Mobility Equipment Research & Development Command
ATTN: DRDME-DS
Fort Belvoir, VA 22060

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:04Z
Check the source to verify that this is the current version before use.